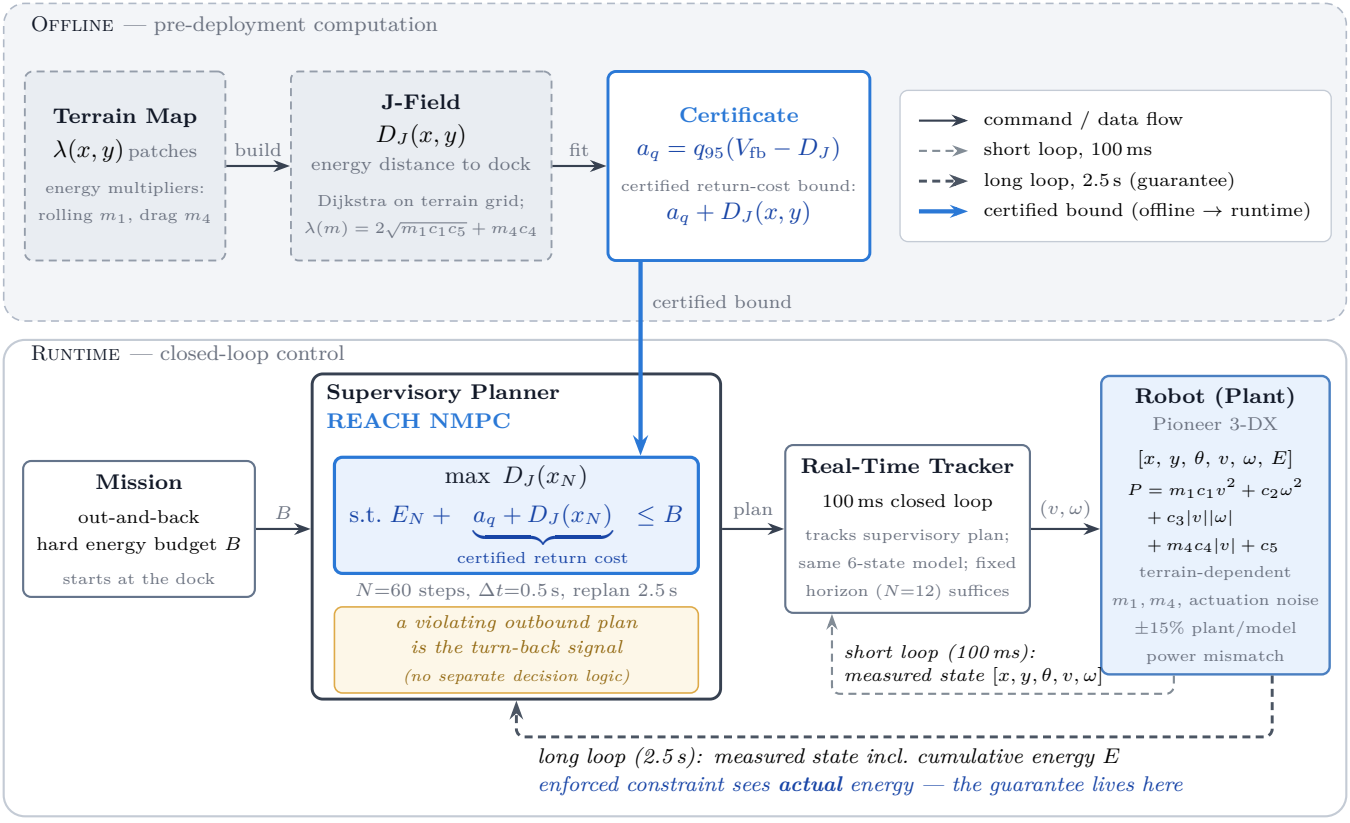

Offline — pre-deployment computation
Terrain Map
$\lambda(x, y)$ patches
energy multipliers:
rolling m_1, drag m_4
build
J-Field
$D_J(x, y)$
energy distance to dock
Dijkstra on terrain grid;
$\lambda(m) = 2\sqrt{m_1 c_1 c_5} + m_4 c_4$
fit
Certificate
$a_q = q_{95}(V_{\text{fb}} - D_J)$
certified return-cost bound:
$a_q + D_J(x, y)$
command / data flow
short loop, 100 ms
long loop, 2.5 s (guarantee)
certified bound (offline → runtime)
certified bound
Runtime — closed-loop control
Supervisory Planner
REACH NMPC
max $D_J(x_N)$
s.t. $E_N + a_q + D_J(x_N) \le B$
certified return cost
N=60 steps, Δt=0.5 s, replan 2.5 s
a violating outbound plan
is the turn-back signal
(no separate decision logic)
Mission
out-and-back
hard energy budget B
starts at the dock
B
plan
Real-Time Tracker
100 ms closed loop
tracks supervisory plan;
same 6-state model; fixed
horizon (N=12) suffices
(v, ω)
Robot (Plant)
Pioneer 3-DX
$[x, y, \theta, v, \omega, E]$
$P = m_1 c_1 v^2 + c_2 \omega^2$
$+ c_3 |v||\omega|$
$+ m_4 c_4 |v| + c_5$
terrain-dependent
m_1, m_4, actuation noise
±15% plant/model
power mismatch
short loop (100 ms):
measured state $[x, y, \theta, v, \omega]$
long loop (2.5 s): measured state incl. cumulative energy E
enforced constraint sees actual energy — the guarantee lives here